

Multi-Stage Ensemble Model

Image Backbone Selection

EfficientNetV2-M, ConvNeXt-Base, and ViT-Base were the three models chosen ,

They were chosen to maximise diversity

Effnet :

Compound scaling of depth, width, and resolution for balanced efficiency

Systematically balances network depth (number of layers), width (channels per layer), and input image resolution to maximize accuracy per FLOP.

Mobile inverted bottleneck (MBConv) An MBConv block starts by expanding a narrow input (few channels) to a higher-dimensional space, applies a depthwise separable convolution (one convolution per channel) to capture spatial features efficiently, and then projects back to a narrow output to reduce computation while maintaining representational power.

ConvNext :

ResNet-style convolutional backbone modernized with Transformer-inspired design choices Expands channels then projects back

down ,increasing representational capacity without excessive compute.

Large convolutional kernels (7×7) and inverted bottlenecks Captures wider spatial context per layer, similar to self-attention's receptive field, enhancing feature aggregation over larger image regions

Layer normalization replacing batch norm Expands channels then projects back down (opposite of classic ResNet), increasing representational capacity without excessive compute.

Vit :

Vision Transformer using self-attention over fixed-size image patches

Divides the image into patches and applies attention to compute pairwise patch relationships, enabling global context modeling from the outset.

Global context modeling across entire image Each token attends to every other token at every layer, capturing long-range dependencies and holistic scene understanding.

Multi-head attention layers for capturing long-range dependencies

Employs multiple parallel attention "heads" to learn diverse feature subspaces simultaneously, improving the model's ability to capture varied patterns.

Diverse Feature Representation

Convolutions specialize in capturing local patterns such as edges, textures, and fine-grained gradients.

Transformers excel at modeling non-local interactions, capturing global spatial relationships, and forming higher-order semantic groupings

Pretrained models are available, Hence optimized implementations are

readily available, enabling fine tuning on them was simply sufficient for good performance

Test-Time Augmentation (TTA)

Each image is transformed into multiple augmented variants at the time of validation

This is done here to reduce the sensitivity to input noise

Also it helps in decreasing overfitting

AutoGluon Tabular Feature Engineering

Categorical Encoding Converts nominal and ordinal variables via one-hot, target, and frequency encodings,

Feature Crossing makes interaction features that capture non linear relations

Imputation and Outlier Handling imputes missing values using model-based or statistical methods and caps extreme outliers to reduce noise

Binning and Scaling Discretizes continuous variables into quantiles or domain-informed buckets, normalizes skewed distributions, and applies transformations like log where beneficial.

Overfitting Control

It also prevents overfitting while generation of large feature sets by Regularized Base Models, Feature Pruning i.e. low importance to features with less information gain and also using cross validation and ensembling

Base-Learner Model

CatBoost

Input: ConvNeXt Features + ViT Features

It is only given Image features and not given any tabular features ,Hence enabling it to predict the most deep image patterns

CatBoost is specifically chosen for this job as

The two models ConvNext and Vit ,even though being architecturally different the conv model builds on going from local feature to global context whereas transformer builds on the global context from the start ,even though there would still be many overlapping features hence making the data to be Noisy cat boost is the perfect model for this noisy data as it uses oblivious tree ,which builds symmetric trees where regularisation is highly prioritised

XgBoost

Input:ViT Features + AutoGluon Tabular Features

The Transformer model gives out highly complex features which are to be

added with the autogluon tabular features

XGBoost's Architecture of level wise tree growth makes it possible to train deeper trees with included regularisation. Hence enabling formation of. highly interactive features making it the best suitable model for this combination.

LightGBM-1 & 2

Input: Effnet/ConvNext features + AutoGluon Tabular Features

Both Effnet and ConvNext are CNNs which have a hierarchical order of detecting features in an image starting from local features like edges, textures to larger patterns, these are to be added to Autogluon tabular features.

LGBM is a model that uses Best first tree growth approach where at any tree the node that gives high impact i.e. largest information gain is split first

This would not go well with the Transformer architecture as the layers there are not ordered hence making it unnecessarily too complex ,whereas due to an ordered nature of convolutions LGBM's efficient architecture could be utilised.

Also OPTUNA was used for 30 trials for automated hyperparameter optimisation in the GBDTs

Meta-Learner

Here a simple model like Logistic Regression is used instead of a complex model like GBDT's as

in a stacking problem the problem of overfitting is prominent ,Hence using a simpler linear model that is not overly flexible does the job right

Here the Logistic Regression model acts as a weighted average for the four base learners